

Unit 2, Lesson 7: Plotting, Ordering, and Comparing Real Numbers

Benchmark

MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers, represented in various forms.

Learning Targets

- ☐ I can compare rational and irrational values.
- ☐ I can order rational and irrational values.
- ☐ I can plot rational and irrational numbers on a number line.

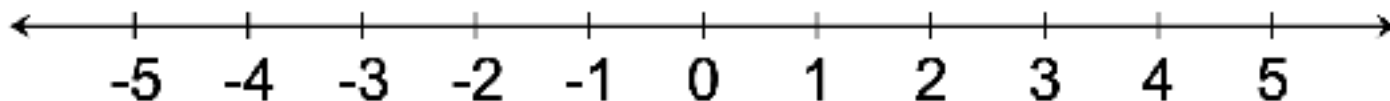
2.7.1 Warm-Up: Bell Work

1. Six real numbers are shown.

$$\sqrt{6}, \quad 12, \quad -3\frac{1}{2}, \quad \frac{4}{3}, \quad -\pi, \quad \sqrt[3]{8}$$

- a. Circle all numbers that are irrational.
- b. Order all the values from greatest to least.

- c. Plot and label each number on the number line.



2.7.2 Activity: Station 1 – Comparing and Ordering

1. Compare each pair of expressions using the symbols $<$, $>$, or $=$.

a. -5π

☐	$<$
☐	$>$
☐	$=$

 $-\sqrt{225}$

b. $\sqrt[3]{125}$

☐	$<$
☐	$>$
☐	$=$

 $\frac{\sqrt{100}}{2}$

c. $\sqrt{45}$

☐	$<$
☐	$>$
☐	$=$

 $\pi + 4$

d. -7

☐	$<$
☐	$>$
☐	$=$

 $-\sqrt[3]{49}$

e. $\frac{29}{2}$

☐	$<$
☐	$>$
☐	$=$

 $\sqrt{80}$

f. $\frac{\pi}{2}$

☐	$<$
☐	$>$
☐	$=$

 $\sqrt{22} - 3$

2. Choose one pair of expressions from the previous question that were not equivalent.

a. Circle the pair of expressions.

b. Explain how you determined which value was greater.

3. Order each set of numbers from least to greatest.

a. $\sqrt{34}$, $\pi + 2$, $5\frac{1}{2}$

b. $-\sqrt[3]{60}$, $-\frac{18}{4}$, $-\pi$

2.7.3 Activity: Station 2 – Plotting Rational and Irrational Numbers on a Number Line

1. Five expressions are shown.

$\sqrt{150}$

$-\pi + 15$

10.2

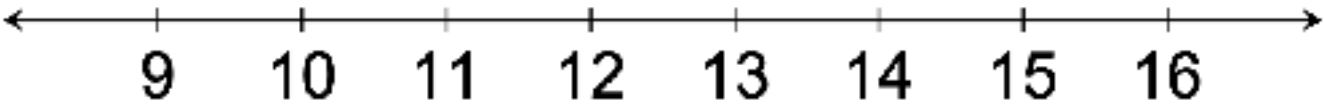
$11\frac{1}{3}$

$\sqrt[3]{64} + 8$

a. Write each expression under the appropriate description in the table.

Includes Only Rational Numbers	Includes an Irrational Number

b. Plot and label each number on the number line.



2. Six expressions are shown.

$-\sqrt{20}$

$-\frac{60}{11}$

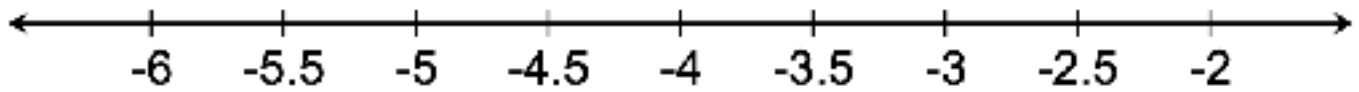
$-\sqrt{8}$

$-\pi$

$\sqrt[3]{-125}$

$-\frac{23}{5}$

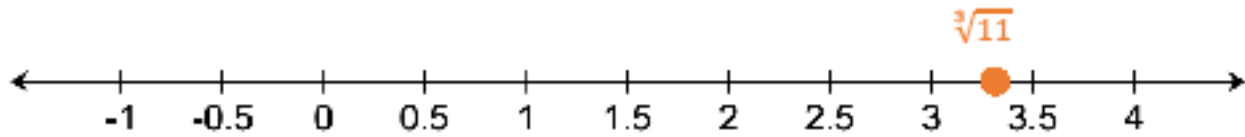
Plot and label the location of each expression on the number line.



2.7.4 Activity: Station 3 – Error Analysis

1. Sara approximated the value of $\sqrt[3]{11}$. Her thinking is shown.

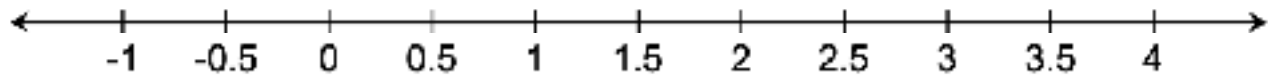
The number 11 is between 9 and 16, which are both perfect squares. So, I know $\sqrt[3]{11}$ lies between 3 and 4 on a number line, but it's closer to 3 because 11 is closer to 9 than 16. I would plot its approximate location here.



a. Discuss Sara’s thinking with your partner. Then, complete the table.

What is correct in Sara’s thinking?	What error(s) did Sara make?

b. Plot and label the correct approximate value of $\sqrt[3]{11}$ on the number line.



2. For each pair of numbers, an explanation comparing the values is provided.

a. Determine whether each explanation is correct or incorrect.

Pair of Numbers	Comparison and Explanation	Is it Correct?
π^2 and 9	9 is greater than π^2 because π is slightly more than 3, so π^2 is slightly more than 6.	☐ correct ☐ incorrect
$\sqrt{50}$ and 8	$\sqrt{50}$ is less than 8 because 8 is equal to $\sqrt{64}$ and $\sqrt{50}$ is less than that.	☐ correct ☐ incorrect
$\frac{13}{2}$ and $\sqrt[3]{60}$	$\frac{13}{2}$ is greater than $\sqrt[3]{60}$ because $\frac{13}{2}$ is slightly more than 4 and $\sqrt[3]{60}$ is slightly less than 4.	☐ correct ☐ incorrect
-2π and -6	-6 is less than -2π because $-2 \cdot \pi$ is between -6 and -7 , which means it's larger than -6 .	☐ correct ☐ incorrect

b. Two of the explanations above are incorrect. Write the correct explanation for each of them by completing the sentence stems.

_____ is greater than _____ because ...

_____ is less than _____ because ...

2.7.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept learned in this unit.



I need help.



I got it.



I could help someone.

Solve equations in the form $x^2 = p$.	  
Solve equations in the form $x^3 = q$.	  
Calculate square roots of perfect squares and cube roots of perfect cubes.	  
Identify whether a number is rational or irrational.	  
Locate the approximate value of expressions involving irrational numbers on a number line.	  
Plot, order, and compare rational and irrational numbers on a number line.	  

